

⊛ Integración de funciones trigonométricas tipo 4:
 $\int R(\sin x, \cos x) dx,$

1) Función impar en $\sin x$: $R(-\sin x, \cos x) =$
 $= - R(\sin x, \cos x).$

cambio: $t = \cos x$

$$t = \cos x$$

$$x = \arccos t$$

$$dx = \frac{-1}{\sqrt{1-t^2}} dt$$

$$\sin x = \sqrt{1-t^2}$$

Ejemplo:

$$\int \frac{\sin x}{1+4\cos^2 x} dx = \overset{\substack{\downarrow \\ t = \cos x}}{-} \int \frac{1}{1+4t^2} dt =$$

$$= - \frac{1}{2} \operatorname{arctg}(2t) + C = \boxed{-\frac{1}{2} \operatorname{arctg}(2\cos x) + C.}$$

2) Función impar en $\cos(x)$: $R(\sin x, -\cos x) =$
 $= -R(\sin x, \cos x)$

Cambio : $t = \sin x$

$$\left. \begin{aligned} t &= \sin x, \\ x &= \arcsin t, \\ dx &= \frac{1}{\sqrt{1-t^2}} dt, \\ \cos x &= \sqrt{1-t^2}. \end{aligned} \right\}$$

Ejemplo:

$$\int \frac{\sin x \cdot \cos x}{\sin x + 1} dx = \int \frac{t}{t+1} dt =$$

$$= 1 - \ln |t+1| + C = \sin x - \ln |\sin x + 1| + C.$$

3) Función por sen $\sin(x)$ y $\cos(x)$:

$$R(-\sin x, -\cos x) = R(\sin x, \cos x)$$

cambio: $\boxed{t = \operatorname{tg}(x)}$

$$t = \operatorname{tg} x$$

$$x = \operatorname{arctg} t$$

$$dx = \frac{dt}{1+t^2}$$

$$\sin^2 x = \frac{t^2}{1+t^2}$$

$$\cos^2 x = \frac{1}{1+t^2}$$

$$dx = \frac{dt}{1+t^2}$$

Ejemplo:

$$\boxed{\operatorname{tg} x = t}$$

$$\int \frac{dx}{1+\cos^2 x}$$

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$$\int \frac{\frac{1}{1+t^2}}{1 + \frac{1}{1+t^2}} dt =$$

$$= \int \frac{1}{t^2 + 2} dt = \frac{1}{\sqrt{2}} \operatorname{arctg} \left(\frac{1}{\sqrt{2}} \operatorname{tg} x \right) + C$$

4) Si no es ninguno de los casos anteriores:

cambio: $\boxed{t = \operatorname{tg} \frac{x}{2}}$

$$\left. \begin{aligned} x &= 2 \operatorname{arctg} t \\ dx &= \frac{2}{1+t^2} dt \end{aligned} \right\}$$

$$\left. \begin{aligned} \cos x &= \frac{1-t^2}{1+t^2} \\ \sin x &= \frac{2t}{1+t^2} \end{aligned} \right\}$$

Ejemplo:

$$\int \frac{1+\cos x}{1-\cos x} dx \stackrel{\downarrow}{=} \int \frac{1+\cos x}{1-\cos x} \frac{2}{1+t^2} dt$$

$$= x - 2 \cotg \left(\frac{x}{2} \right) + C$$